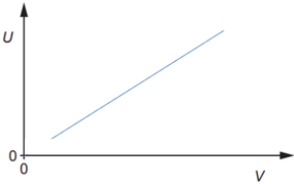


	internal energy decreases	[1]
3(a)	molecule(s) rebound from wall of vessel / hits walls/ collide change in momentum gives rise to impulse / force average change in momentum/impulse gives rise to the average (constant) pressure/force.	[1] [1] [1]
(b)	$p = \frac{1}{3} \rho \langle c^2 \rangle$ $1.02 \times 10^5 = \frac{1}{3} \times 0.900 \times \langle c^2 \rangle$ $\langle c^2 \rangle = 3.4 \times 10^5$ $c_{\text{rms}} = 580 \text{ m s}^{-1}$	[1] [1]
(c)	Straight line with positive gradient Line passing through origin (can never reduce the volume to zero thus the graph need not be shown passing through the origin) 	[1]

3

4(a)	amplitude = 0.50 cm	[1]
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